

Statistics

2025 Lectures

Part 1 - Introduction

Institute of Economic Studies
Faculty of Social Sciences
Charles University in Prague

Outline

1. Summary of Introductory Statistics
2. Motivation for Statistics
3. Syllabus of Statistics
4. Requirements

Summary of Introductory Statistics

- data set, sample statistics
- summary and visualization of data
- sample space, events
- definitions of probability
- counting techniques
- conditional probability, Bayes theorem, independence of events
- sampling techniques

Why Statistics (for Economists)?

- **Decisions under uncertainty:** policy, pricing, hiring, investment.
- **From models to evidence:** data confronts theory; theory guides what to measure.
- **Separate signal from noise:** what is real vs. random variation?

Core idea

Statistics is the logic of uncertainty: a disciplined way to learn about the world from incomplete and noisy data.

Mindset

Two quick stories

Policy A vs. Policy B (jobs):

- Observational data: Policy A regions show higher employment, is it causal or selection?
- Statistical tools: design, controls, uncertainty quantification.

Market move:

- Equity index drops by 1.2% today, is it a rare shock or typical fluctuation?
- We need models of variation to interpret “surprises.”

What you'll be able to do

- **Quantify uncertainty:** distributions, expectations, variability.
- **Make principled comparisons:** estimation, confidence, testing.
- **Link models to data:** regression, prediction, causal questions.

Mindset

Intuition $\rightarrow$ formal definition $\rightarrow$ computation $\rightarrow$ interpretation.

How we'll work (expectations)

- We value clear reasoning and clean notation more than memorized formulas.
- Examples first, then the formal statement and proof idea.
- We will always ask: What does this number mean for an economic decision?

Statistics: People and times

- **Lecturer:** Luboš Hanus
 - email: lubos.hanus@fsv.cuni.cz
 - web: IES
- **Course link:**
- **Lectures:** Monday, 9:30, lecture hall O109
- **Office/Consultation hours:** Monday, 13:30-15:00, room 507,
by appointment via email.
- **Seminars and TAs:** Mon 11:00 [Josef - O314], Mon 15:30 [Luboš/Ying - O314], Tue 9:30 [Matěj - O109]

Course material

Material:

- Lecture handouts
- Seminar handouts + R codes
- Solution to home assignments

Methods:

- 'pen and paper'
- R and RStudio (free)

Syllabus

Week 1 29. 9. 2025 → Introduction + Random Variables

Week 2 6. 10. 2025 → Random Variables

Week 3 13. 10. 2025 → Multivariate Distributions

Week 4 20. 10. 2025 → Expectations

Week 5 ~~27. 10. 2025~~ → Dean's day

~~28. 10. 2025~~ → Independent Czechoslovak State Day

Week 6 3. 11. 2025 → Conditional Distributions and Expectation

Syllabus

Week 7 10. 11. 2025 → Selected Families of Distribution

Week 8 ~~17. 11. 2025~~ → *No seminars*

18. 11. 2025 → Random Samples

Week 9 24. 11. 2025 → Midterm

Week 10 1. 12. 2025 → Point Estimation

Week 11 8. 12. 2025 → Interval Estimation

Week 12 15. 12. 2025 → Testing statistical hypotheses

Week 13 5. 1. 2026 → Additional lecture if needed (TBA)
or any other date...

Requirements

- Midterm test – 20 points – November 24th
 - Homeworks – 20 points – Five assignments:
 - W03 4 points, W05 4 points, W07 4 points, W10 4 points, W12 4 points + extra activity replacing one lowest score
 - Exam test – 60 points; at least 30.5 points or more, else F
 - minimum requirement of more than 50% from the exam test
 - Book reading + Active work
- A 91% – 100%
- B 81% – 90%
- C 71% – 80%
- D 61% – 70%
- E 51% – 60%
- F 0% – 50%

Time Requirements

- The credit load of 7 ECTS corresponding to 175 hours $\approx$ 7x25 or students' work:
 - participation lecture time – 16 hours
 - participation seminar time – 16 hours
 - time anticipated for work on homeassignments – 5x12 hours
 - time anticipated for midterm preparation – 20 hours
 - time anticipated for exam preparation – 45 hours
 - exam – 3 hours
 - time anticipated for additional home study and reading during the semester – 15 hours

Exam Dates

- No exam dates in 2025!
- Exam dates for 2026 will be announced during the semester, no later than December 8th, 2025.

Literature

Mandatory **book reading** and main source of problems for **seminar class and home assignments**:

- BARTOSZYNSKI, R. and NIEWIADOMSKA-BUGAJ, M. Probability and Statistical Inference. 3rd ed, Wiley, 2021 (Sections 5.1 – 12.9)
- RAMACHANDRAN, K.M., TSOKOS, C.P. Mathematical Statistics with Applications in R, 3rd ed, Elsevier, 2020 (Sections 2.5 – 6.6)

Further recommended but only voluntary:

- ANDERSON D.R. et al. Statistics for Business and Economics. 15th ed, Cengage Learning, 2023
- MITTELHAMMER, R.C. Mathematical Statistics for Economics and Business. 2nd ed, Springer, 2013
- SUHOV, Y., KELBERT, M. Probability and Statistics by Example, volume 1, 2nd ed Cambridge Uni Press, 2014.